

FloodRisk FYRE — User Guide

A 3D flood-resilience game for a riverside campus. Make decisions, watch the flood rise, see what fails.

► [Play: final-eight-theta-85.vercel.app](https://final-eight-theta-85.vercel.app)

1. Open & basic controls

1. Open the link above in a desktop browser (Chrome, Safari, Edge, Firefox).
2. On the start screen, choose one of the three modes below.
3. In any 3D view: **drag** = orbit · **scroll / pinch** = zoom · **click a building** = scan it (damage, flood depth, power, loss).
4. To exit a mode, click **×** (top-left) or the **≈ logo** to return to the menu.

Reading the campus: lit windows = powered, dark windows = no power · cyan power lines = energized, red lines = failed · a red ⚡ **OUT / DARK** tag over a building means it lost power (DARK = dry but still lost power).

2. Live Drill ⌚ TIMED + SCORE

A timed emergency. The clock runs, the river rises, you act before deadlines.

1. **Prep:** choose storm severity (Minor → Extreme) and tick the capital upgrades to buy up front (drainage, elevate buildings, raise substation, elevate roads). Stay within budget.
2. Click **Start the drill**. The clock begins and water rises.
3. **Answer the alert cards** (deploy barriers, evacuate, sandbag the substation, close roads, open a shelter). Each has a shrinking timer — act before it runs out, or the option is lost. Acting **earlier** helps more.
4. Track the top bar: clock, water level, loss, people at risk, buildings dark, budget.
5. At the end you get a **resilience score (0–100) and grade**, broken down by loss avoided, people protected, lifelines, and budget.
6. Use **Re-run drill** to try other choices, or **Back to menu**.

Goal: highest score = lowest loss, most people protected, power kept on, within budget.

3. Decision Lab 🔄 COMPARE A vs B

Two tabs at top-right: **Guided dilemmas** and **Free compare**.

Guided dilemmas

1. Read the situation and the two plans (A and B).
2. Click a plan chip to view that plan's campus in 3D; click the other to flip.
3. **Predict:** click the plan you think is more resilient.
4. See both outcomes revealed side-by-side, plus the concept it teaches.
5. **Next dilemma** to continue, or **Retry**.

Free compare

1. Set **Plan A** (left) and **Plan B** (right): a flood scenario + any mitigations.
2. Click a panel header (**view in 3D**) to show that plan's campus.
3. Read the **scoreboard**: damage, buildings dark, people, downtime, spend, cost of risk. The better value in each row is highlighted, with a recommended plan and one-line insight.

4. Sandbox 💎 FREE PLAY

1. Pick a **flood scenario** (top-right).
2. Toggle **mitigations** (left) — the 3D campus and CAT dashboard update instantly.
3. Switch **role** (Emergency Manager, Engineer, Budget Officer, Risk Analyst) to change what the dashboard emphasizes.
4. **Click any building** to scan its details.
5. Click **Run before / after reflection** to compare your plan against doing nothing.

Risk = Hazard × Exposure × Vulnerability

You can't change the hazard (the flood), but you **can** change exposure and vulnerability. Watch for the **cascade**: if the power substation floods, every building it feeds goes dark — even dry ones on high ground. Often, protecting the lifeline (raising the substation, lowering the water) does more for resilience than reducing flood damage to individual buildings.